

Anime-Like Motion Transfer with Optimal Viewpoints

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ABSTRACT

In 3D character animation, the frame rate is often reduced to mimic hand-drawn animations, like anime (Japanese animation). However, anime with a low frame rate differs from real motions because it omits excessive movements and emphasizes speed by expressing motions with a small number of impressive poses. It is difficult to reproduce such motions only by downsampling mocap data. Thus, in this poster, we propose a method for converting mocap data into anime-like 2D motions by respecting production site techniques. The proposed method evaluates the characteristics of motion data using the time distributions of speeds and pose areas to select an appropriate sequence of viewpoints and extract effective poses for each viewpoint.

CCS CONCEPTS

• Computing methodologies → Procedural animation.

KEYWORDS

3D character, anime, motion style transfer

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1 INTRODUCTION

Animating on ones is a common process that incorporates 24 drawings per second. In hand-drawn animation, each picture is held for two or three frames, resulting in 12 or 8 frames per second (fps), referred to as animating on twos and threes, respectively. These are preferred styles in anime (Japanese animation). In anime, *nakawari* has become a well-established technique that omits the superfluousness inherent in realistic motions and incorporates considerable changes in speed to make the character's motions more visually attractive.

Current 3DCG character animation often uses downsampling to mimic hand-drawn anime-like motion expressions. However, sampling real motions at equal intervals does not produce the same effect as motions generated by *nakawari*. Manual editing also requires artistic experience and much hard work. In this poster, therefore, we propose a method for converting mocap data into

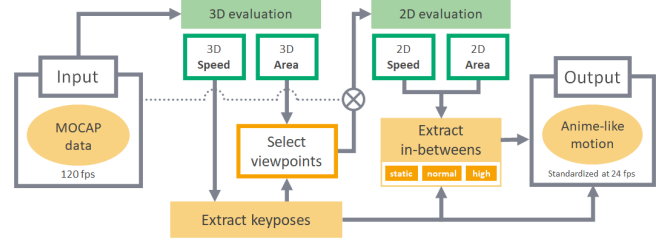


Figure 1: Basic architecture of the proposed framework.

anime-like motions based on the *nakawari* technique. To evaluate motion characteristics quantitatively, we focus on the time distributions of speeds and pose areas of a human character, and we simultaneously select effective viewpoints and extract poses for each viewpoint.

2 METHOD

Figure 1 shows the basic architecture of the proposed framework, where the time distributions of speeds and the pose areas are computed to evaluate the dynamism of a target motion data. The motion speed of a frame is derived from the sum of the differences between each of the coordinates of 12 joints (shoulders, arms, uplegs, and legs) in 3D space in the adjacent frame. The poses corresponding to the local minima are extracted as keyposes [Miura et al. 2014]. The shape of a pose is approximated by a triangle to evaluate the area of the silhouette quantitatively. We chose the triangle, as it has the greatest area with three vertices out of 10 joints, such as the head, hip, arms, and legs.

We designed an algorithm to determine the timings for switching and to select viewpoints for each of the time segments. The best viewpoint for a segment is selected by maximizing the sum of the areas of all frames when the pose triangle in 3D space is projected onto the 2D screen. The viewpoint for each of the frame intervals partitioned by keyposes is searched in turn, and if the viewpoint switch is less than 30 degrees, the segments are merged and the viewpoint is reselected. If the integrated segment exceeds a certain number of seconds, the viewpoint is forcibly switched. The camera position of the new cut is restricted below the “180-degree rule” so it does not end up behind the character.

In-betweens are extracted for animating on threes according to the speeds between each pair of consecutive keyposes. Because the motion is seen differently depending on the viewpoint, the speeds and pose areas are computed from the 2D projected coordinates. Further, because the frames around the keyposes are prone to including extra movements inherent in real movements, as opposed to anime movements, an in-between is not extracted until sufficient

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Figure 2: A sequence of extracted poses for pitching motion using the proposed method. The circle marks represent the keyposes and the diamond mark represents the pose animated on twos. The dashed triangles indicate pose areas on a 2D screen.

pose differences are detected, and a pose with the minimum speed is shown instead.

Distance relationships among poses in nakawari can be classified into two categories: those that divide the distance by a constant and those that include easing. We propose a single nakawari-based pose extraction algorithm that can output both equidistant and easing motions. First, the poses with extremal pose areas are extracted to divide the segment into shorter ones. Next, the consecutive frames whose sum of the speeds is the largest in terms of the number of frames are identified by calculating the number of possible drawings in each segment, and the remaining poses are selected so the distances are eased toward both ends of the frames.

If the maximum speed between keyposes exceeds a threshold value, the segment is considered fast motion, and in-betweens are extracted to emphasize the speed. In anime, fast motions can be expressed by omitting intermediate motions intentionally [Kitamura et al. 2013]. By omitting the frames until the 3D speeds fall below a threshold value, the motion is emphasized as faster than that in reality. At this time, a specific pose from the omitted high-speed motion is animated on twos to reduce the interruption of the motion due to the omission. It also slows down the un-omitted motions to fill between the omitted frames to emphasize fast motion relatively.

3 RESULTS AND EVALUATION

The output motions can be seen in Fig. 2 and the supplementary video. The viewpoint selection algorithm optimized the location of the viewpoint to provide high visibility of the 3D pose area, so the motion can be viewed from the front. In-betweens were selected to prioritize frames with extremal pose area values and to avoid frames with high speeds, thus achieving a nakawari-based and slow-in/slow-out motion.

We quantitatively evaluated the motions and viewpoints generated by the proposed method using a pairwise comparison with 8-fps downsampling motions or fixed viewpoints. The evaluation adopted a scoring system where a score gets closer to 10 if the proposed method is more attractive. The order of display and labeling of motion types were randomly changed for each participant, and responses were obtained from nine students in the authors' laboratories. Figure 3 shows the results of the evaluation. Meanwhile, because real motion differs slightly, even in periodic motions, our method's emphasis on speed did not function in a unified manner, and speed was inconsistent. Some participants commented that the monotonous background made it difficult to identify the positions of the characters after switching viewpoints.

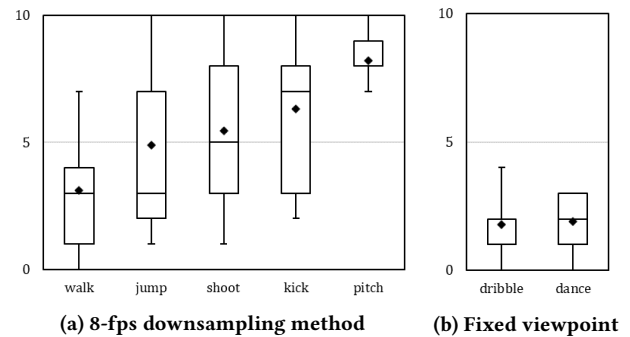


Figure 3: Comparison scores of the proposed method to 8-fps downsampling motions or fixed viewpoints.

4 CONCLUDING NOTES

In this study, we proposed a method of reproducing anime-like motions by extracting poses from mocap data based on the time distributions of speeds and pose areas. Motion transfer based on the nakawari method can relieve the redundancy caused by the physical speed constraints of realistic motions and can effectively extract poses more suitable for a lower frame rate than the uniform downsampling method.

Future work should consider a method to generate new poses not included in the original motion data, which would allow for more exaggerated and thus more anime-like motions. We also believe enhancing the interface to allow the user to specify poses and shots would allow for highly interactive maneuverability.

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